Reading Passage: The Evolution of Artificial Intelligence

Artificial Intelligence (AI) **has transformed** from a theoretical concept into one of the most powerful engines of modern technology. Although ancient mythologies **were already describing** mechanical beings centuries ago, the formal field of AI research **was born** in **1956** at the historic Dartmouth Conference. While researchers **were discussing** the future of computing during that summer, John McCarthy **coined** the term "Artificial Intelligence."

In its early decades, AI research **relied** heavily on Symbolic AI. While computers **were running** early logic programs, engineers **were manually coding** thousands of explicit rules into software. However, these early systems **struggled** whenever they **encountered** real-world noise and ambiguity. While scientists **were promising** rapid breakthroughs, progress **was slowing** down due to hardware limitations. Because researchers **had failed** to deliver on their original promises, investors **withdrew** their financial support. As a result, the industry **entered** severe periods of reduced funding known as **"AI Winters."**

The major turning point **occurred** in the late 1990s and early 2000s while computer scientists **were developing** advanced statistical methods. Instead of writing fixed rules, engineers **began** feeding massive datasets into algorithms. While chess grandmaster Garry Kasparov **was competing** against IBM's **Deep Blue** in 1997, the supercomputer **was processing** millions of potential moves per second. Deep Blue **defeated** Kasparov, proving that machines **had achieved** master-level strategic calculation.

By the early 2010s, a dramatic breakthrough **took** place. While tech companies **were collecting** unprecedented amounts of internet data, researchers **realized** that multi-layered artificial neural networks **could process** complex visual and linguistic patterns. In 2012, while competitors **were using** traditional algorithms in the ImageNet competition, the **AlexNet** model **won** by a massive margin using deep learning on GPUs. Since the introduction of the **Transformer architecture** in 2017, AI systems **have evolved** rapidly, leading to the generative models that **are changing** the world today.

I. Grammar Identification (Past Tenses in Context)

1. **Past Continuous (Simultaneous Actions):** Locate the sentence in Paragraph 3 describing what Garry Kasparov and Deep Blue were doing at the same time. Write the two verb forms used:
   - *Kasparov:* _____
   - *Deep Blue:* _____
2. **Past Simple vs. Past Continuous (Interrupted Action):** Complete the sentence using the correct form of the verbs in brackets based on the text:
   - While researchers _____ (*promise*) rapid breakthroughs, progress _____ (*slow*) down.

Part I: Multiple Choice

Choose the best answer for each question.

1. **In which year was the term "Artificial Intelligence" formally coined?**
   - A) 1945
   - B) 1956
   - C) 1997
   - D) 2012
2. **What was the primary methodology used during the early decades of AI research?**
   - A) Deep Learning
   - B) Statistical Modeling
   - C) Symbolic AI
   - D) Neural Networks
3. **What does the term "AI Winter" refer to in the context of the text?**
   - A) A period where AI was used to predict weather patterns.
   - B) The technical limitations of running computers in cold climates.
   - C) Eras of reduced funding and interest due to unmet expectations.
   - D) The initial discovery of AI in a laboratory during winter.
4. **What was the significance of the match between Garry Kasparov and Deep Blue?**
   - A) It was the first time a computer played a game of chess.
   - B) It proved that machines could achieve master-level strategic calculation.
   - C) It led to the immediate creation of the internet.
   - D) It demonstrated that AI could feel human emotions.
5. **Which technology was crucial for the success of the AlexNet model in 2012?**
   - A) Symbolic Logic

- B) Manual coding of rules
- C) Deep learning on GPUs
- D) Ancient mechanical blueprints

Part II: Short Answer

Answer the following questions based on the details in the text.

1. **According to the text, why did early AI systems struggle with real-world applications?**
2. **What major shift in methodology occurred during the late** 1990**s and early** 2000**s?**
3. **What architecture, introduced in** 2017**, paved the way for modern generative models?**